Cloud Computing for Humanities and Social Sciences

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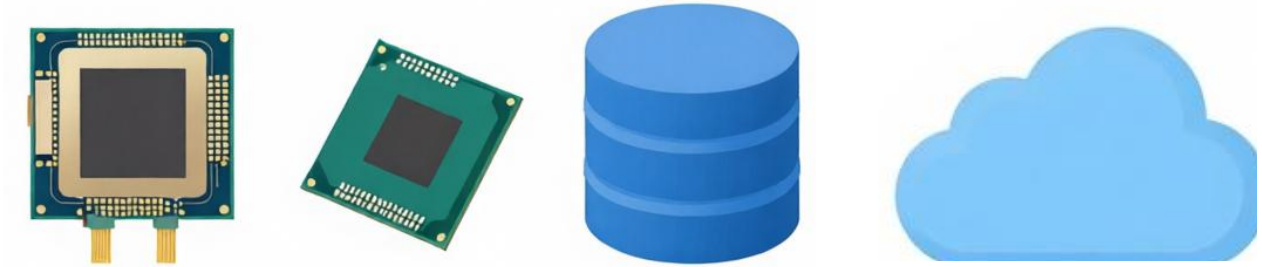
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Presentation outline.

- What cloud computing is and why researchers use it
- Cloud types (public, private, hybrid)
- An overview of the Digital Research Alliance of Canada's cloud options
- When to choose cloud computing over HPC
- Common research use cases and practical considerations
- Demonstration of launching a cloud virtual machine and hosting a basic website

What is Cloud Computing?

Delivery of computing resources and services over the internet



Running OCR

Processing transcripts

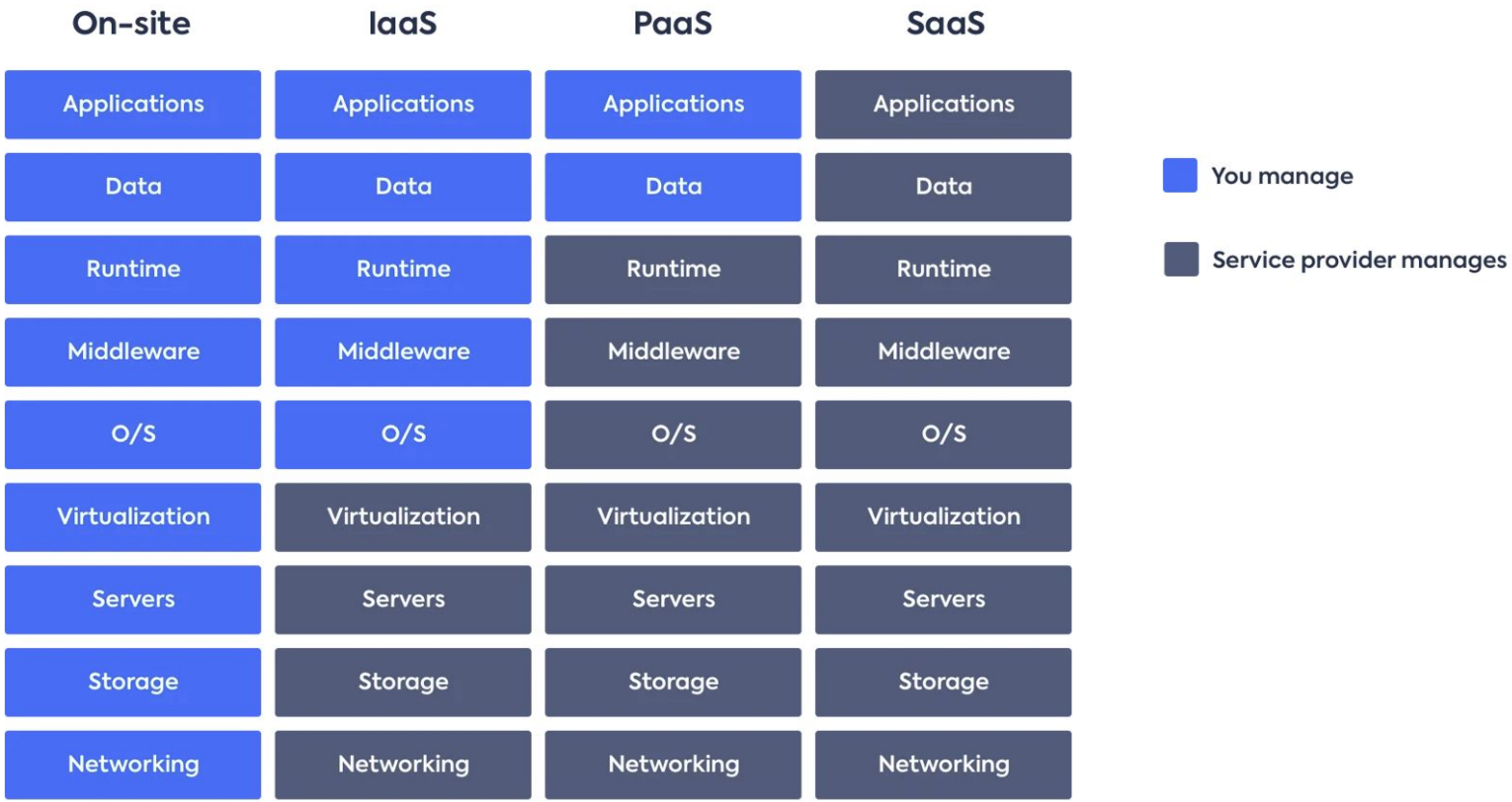
Object Detection

Image analysis

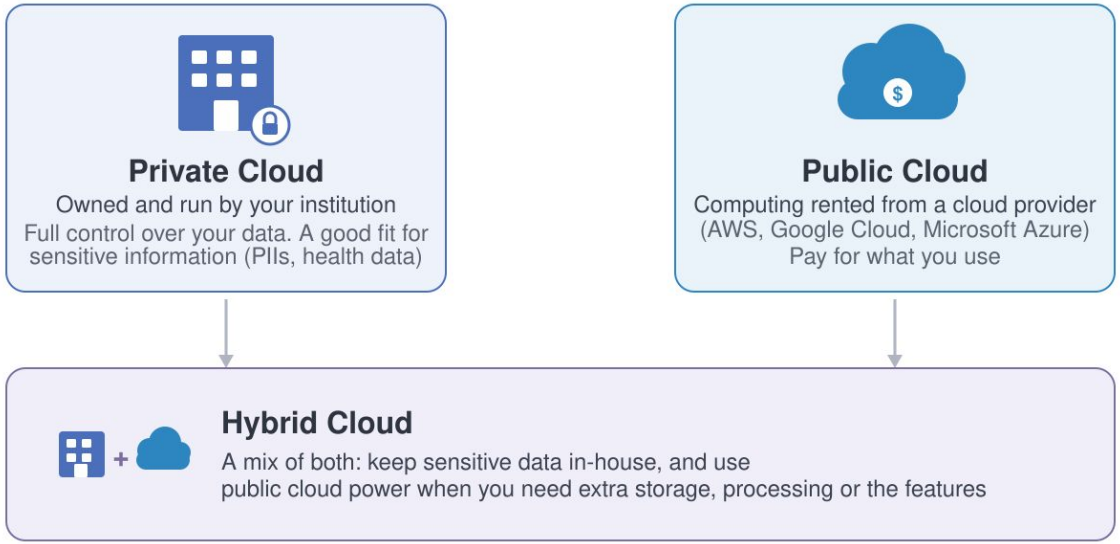
Research Data Storage

Digital Archive

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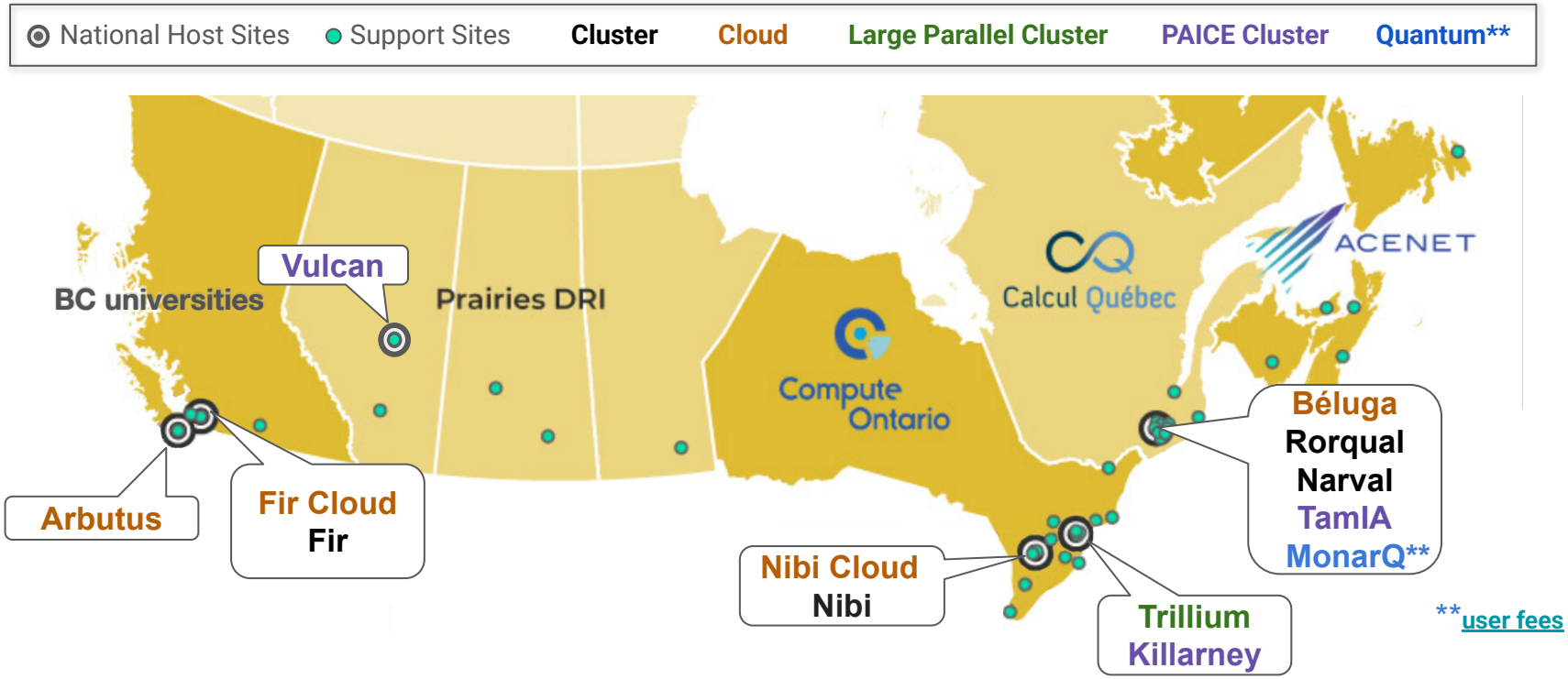

On-site
IaaS
PaaS
SaaS
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
You manage
Service provider manages

Types of Cloud Computing.



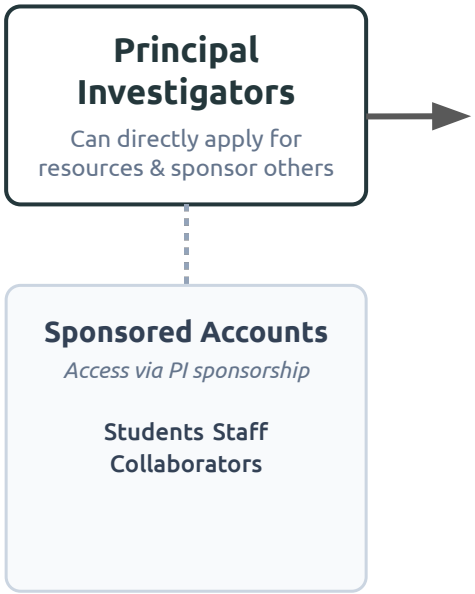
Canada's Advanced Research Computing Platform.

Digital Research Alliance of Canada



National Systems | PAICE Systems

Accessing Alliance Cloud resources.



NON-COMPETITIVE

Rapid Access Service (RAS)

Alliance Account Setup

Request access through CCDB

Fill out the RAS request form

COMPETITIVE

Resource Allocation Competition (RAC)

I need more than RAS

Eligible roles: Faculty, Adjunct Faculty, and Librarian positions only.

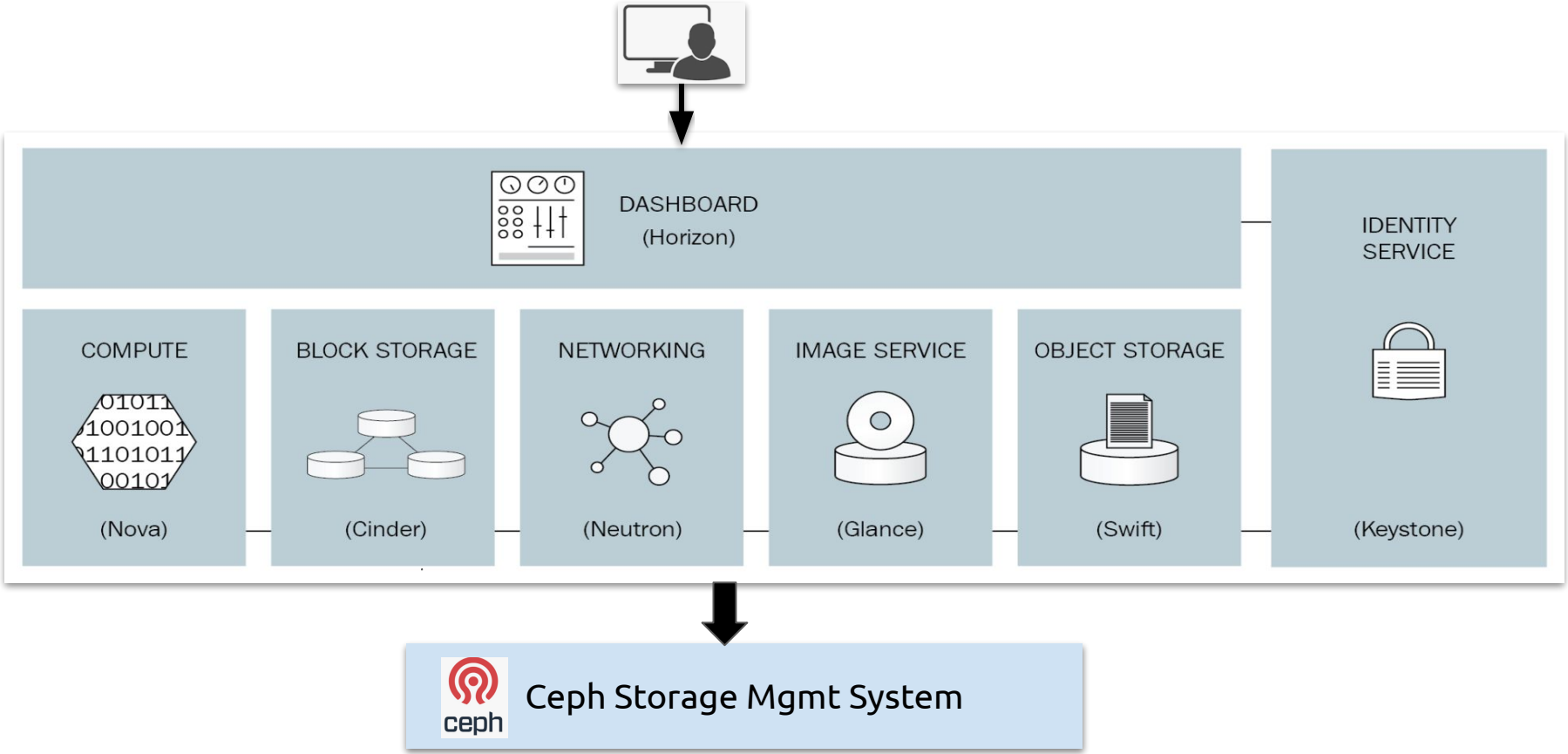
Any **available resource** can be requested.

No maximum limit imposed on resource requests.

24Gb vGPU flavor

RAC application guide

Private Cloud: Openstack-based Alliance cloud (Arbutus).



Alliance Cloud: Instance Types.

Compute Instances

Short-to-medium term workloads.

- **Max Wall-Time:** 1 month*
- **Use Case:** High CPU, Memory flavors
- **RAS Quota:** Up to 80 vCPUs / 300GB RAM

Persistent Instances

Long-running persistent services

- **Duration:** Runs indefinitely
- **Use Case:** Web servers, database servers
- **RAS Quota:** Up to 25 vCPUs / 50GB RAM

More at: Cloud RAS resource limits

* might be enforced in the future



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Alliance Cloud: Storage Types (upto 10 TB each for RAS).



Persistent Volume Storage (Block)

- High-performance block storage attached directly to VMs
- Supports persistent OS boot disks and volume snapshots



Object Storage (S3)

- Storage for unstructured data
- Accessible via HTTP API (offered on **Arbutus** cloud)



Shared Filesystem (Ceph)

- Shared file storage attachable to multiple instances simultaneously
- Ideal for collaborative research (offered on **Arbutus** cloud)

	Cloud	HPC
Use Cases	Web hosting, persistent services, interactive data analysis, custom environments	Massive parallel computations, complex simulations, high-performance batch modeling
Environment & Control	Full root access, flexible Virtual Machine hardware, flexible OS (including Windows), automated deployment	Pre-configured multi user environment, module system, non-root user permissions
Execution Model	Always available persistent servers, instant interactive access	Batch job submission via scheduler (Slurm), queued execution, time-bound tasks
Hardware & Network	Virtualized hardware instances (vCPUs, vGPUs), standard ethernet network connectivity	Multi CPU/GPU nodes with ultra-fast, low-latency inter communication

Example Cloud Projects.

Tools hosted on Arbutus cloud:

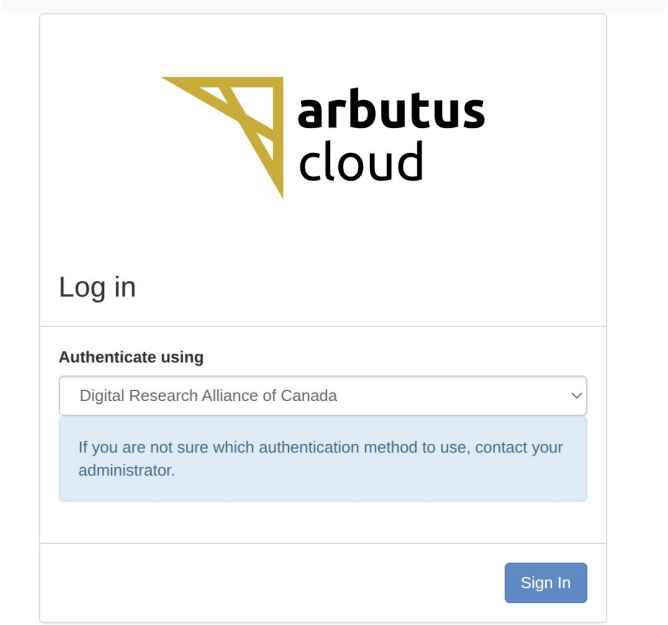
- CSSL Weather Intelligence
- Umple.org
- Voyant
- Surveys

Hostname format to setup SSL certs: xxx-xxx-xxx-xxx.arbutus.alliancecan.ca

Demo - Arbutus cloud.

Objectives

- Deploy a VM on Openstack https://arbutus.alliancecan.ca/
- Configure an application on the VM to serve web pages
- Create a Object Storage bucket and place an image into it
- Use the image in the Object Storage bucket on a web page
- Use HTTPS



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Thank you!

Questions?

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Useful Links.

- Cloud RAS allocation
- Cloud quick start
- Short videos:
  - Introduction to cloud
  - Accessing Arbutus cloud
- RAC application guide

Useful Commands.

```
sudo apt-get install apache2 -y
cat > /var/www/html/index.html << EOF
<!DOCTYPE html>
<html lang="en">
<head>
   <title>My First Website</title>
</head>
<body>
   <h1>Welcome to My Website!</h1>
   <p>This is my first webpage.</p>
   <h2>About</h2>
   <p>This is where I will share information about my website.</p>
   <!-- <img src="image.jpg" alt="Description of image"> -->
</body>
</html>
EOFa
sudo service apache2 restart
sudo snap install --classic certbot
sudo certbot --apache
```

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